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Java SE 27 & JDK 27

Class Buffer

java.lang.Object
    java.nio.Buffer

Direct Known Subclasses:

ByteBuffer, CharBuffer, DoubleBuffer, FloatBuffer, IntBuffer, LongBuffer, ShortBuffer

---

```
public abstract sealed class Buffer
extends Object
permits ByteBuffer, CharBuffer, DoubleBuffer, FloatBuffer, IntBuffer,
LongBuffer, ShortBuffer
```

A container for data of a specific primitive type.

A buffer is a linear, finite sequence of elements of a specific primitive type. Aside from its content, the essential properties of a buffer are its capacity, limit, and position:

> A buffer's *capacity* is the number of elements it contains. The capacity of a buffer is never negative and never changes.
>
> A buffer's *limit* is the index of the first element that should not be read or written. A buffer's limit is never negative and is never greater than its capacity.
>
> A buffer's *position* is the index of the next element to be read or written. A buffer's position is never negative and is never greater than its limit.

There is one subclass of this class for each non-boolean primitive type.

Transferring data

Each subclass of this class defines two categories of *get* and *put* operations:

> *Relative* operations read or write one or more elements starting at the current position and then increment the position by the number of elements transferred. If the requested transfer exceeds the limit then a relative *get* operation throws a `BufferUnderflowException` and a relative *put* operation throws a `BufferOverflowException`; in either case, no data is transferred.
>
> *Absolute* operations take an explicit element index and do not affect the position. Absolute *get* and *put* operations throw an `IndexOutOfBoundsException` if the index argument exceeds the limit.

Data may also, of course, be transferred in to or out of a buffer by the I/O operations of an appropriate channel, which are always relative to the current position.

Marking and resetting

A buffer's *mark* is the index to which its position will be reset when the `reset` method is invoked. The mark is not always defined, but when it is defined it is never negative and is never greater than the position. If the mark is defined then it is discarded when the position or the limit is adjusted to a value smaller than the mark. If the mark is not defined then invoking the `reset` method causes an `InvalidMarkException` to be thrown.

Invariants

The following invariant holds for the mark, position, limit, and capacity values:

> `0` <= *mark* <= *position* <= *limit* <= *capacity*

A newly-created buffer always has a position of zero and a mark that is undefined. The initial limit may be zero, or it may be some other value that depends upon the type of the buffer and the manner in which it is constructed. Each element of a newly-allocated buffer is initialized to zero.

Additional operations

In addition to methods for accessing the position, limit, and capacity values and for marking and resetting, this class also defines the following operations upon buffers:

- `clear()` makes a buffer ready for a new sequence of channel-read or relative *put* operations: It sets the limit to the capacity and the position to zero.
- `flip()` makes a buffer ready for a new sequence of channel-write or relative *get* operations: It sets the limit to the current position and then sets the position to zero.
- `rewind()` makes a buffer ready for re-reading the data that it already contains: It leaves the limit unchanged and sets the position to zero.
- The `slice()` and `slice(index,length)` methods create a subsequence of a buffer: They leave the limit and the position unchanged.
- `duplicate()` creates a shallow copy of a buffer: It leaves the limit and the position unchanged.

Read-only buffers

Every buffer is readable, but not every buffer is writable. The mutation methods of each buffer class are specified as *optional operations* that will throw a `ReadOnlyBufferException` when invoked upon a read-only buffer. A read-only buffer does not allow its content to be changed, but its mark, position, and limit values are mutable. Whether or not a buffer is read-only may be determined by invoking its `isReadOnly` method.

Thread safety

Buffers are not safe for use by multiple concurrent threads. If a buffer is to be used by more than one thread then access to the buffer should be controlled by appropriate synchronization.

Invocation chaining

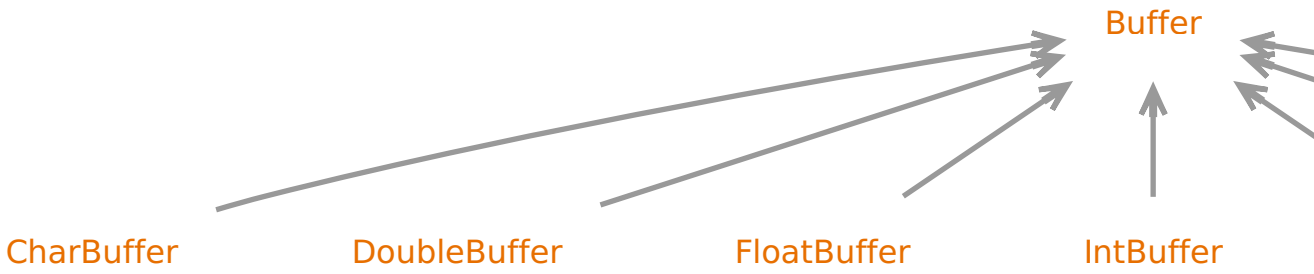
Methods in this class that do not otherwise have a value to return are specified to return the buffer upon which they are invoked. This allows method invocations to be chained; for example, the sequence of statements

```
    b.flip();
    b.position(23);
    b.limit(42);
```

can be replaced by the single, more compact statement

```
    b.flip().position(23).limit(42);
```

Sealed Class Hierarchy Graph:



Since:

1.4

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
abstract Object	**array**()	Returns the array that backs this buffer *(optional operation)*.
abstract int	**arrayOffset**()	Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.
final int	**capacity**()	Returns this buffer's capacity.
Buffer	**clear**()	Clears this buffer.
abstract Buffer	**duplicate**()	Creates a new buffer that shares this buffer's content.
Buffer	**flip**()	Flips this buffer.
abstract boolean	**hasArray**()	

		Tells whether or not this buffer is backed by an accessible array.
final boolean	**hasRemaining**()	Tells whether there are any elements between the current position and the limit.
abstract boolean	**isDirect**()	Tells whether or not this buffer is *direct*.
abstract boolean	**isReadOnly**()	Tells whether or not this buffer is read-only.
final int	**limit**()	Returns this buffer's limit.
Buffer	**limit**(int newLimit)	Sets this buffer's limit.
Buffer	**mark**()	Sets this buffer's mark at its position.
final int	**position**()	Returns this buffer's position.
Buffer	**position** (int newPosition)	Sets this buffer's position.
final int	**remaining**()	Returns the number of elements between the current position and the limit.
Buffer	**reset**()	Resets this buffer's position to the previously-marked position.
Buffer	**rewind**()	Rewinds this buffer.
abstract Buffer	**slice**()	Creates a new buffer whose content is a shared subsequence of this buffer's content.
abstract Buffer		Creates a new buffer whose content is a

	slice(int index, int length)	shared subsequence of this buffer's content.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Details

capacity

`public final int capacity()`

Returns this buffer's capacity.

Returns:

The capacity of this buffer

position

`public final int position()`

Returns this buffer's position.

Returns:

The position of this buffer

position

`public Buffer position(int newPosition)`

Sets this buffer's position. If the mark is defined and larger than the new position then it is discarded.

Parameters:

`newPosition` - The new position value; must be non-negative and no larger than the current limit

Returns:

This buffer

Throws:

`IllegalArgumentException` - If the preconditions on `newPosition` do not hold

limit

```
public final int limit()
```

Returns this buffer's limit.

Returns:

The limit of this buffer

limit

```
public Buffer limit(int newLimit)
```

Sets this buffer's limit. If the position is larger than the new limit then it is set to the new limit. If the mark is defined and larger than the new limit then it is discarded.

Parameters:

`newLimit` - The new limit value; must be non-negative and no larger than this buffer's capacity

Returns:

This buffer

Throws:

`IllegalArgumentException` - If the preconditions on `newLimit` do not hold

mark

```
public Buffer mark()
```

Sets this buffer's mark at its position.

Returns:

This buffer

reset

```
public Buffer reset()
```

Resets this buffer's position to the previously-marked position.

Invoking this method neither changes nor discards the mark's value.

Returns:

This buffer

Throws:

`InvalidMarkException` - If the mark has not been set

clear

```
public Buffer clear()
```

Clears this buffer. The position is set to zero, the limit is set to the capacity, and the mark is discarded.

Invoke this method before using a sequence of channel-read or *put* operations to fill this buffer. For example:

```
    buf.clear();     // Prepare buffer for reading
    in.read(buf);    // Read data
```

This method does not actually erase the data in the buffer, but it is named as if it did because it will most often be used in situations in which that might as well be the case.

Returns:

This buffer

flip

```
public Buffer flip()
```

Flips this buffer. The limit is set to the current position and then the position is set to zero. If the mark is defined then it is discarded.

After a sequence of channel-read or *put* operations, invoke this method to prepare for a sequence of channel-write or relative *get* operations. For example:

```
    buf.put(magic);    // Prepend header
    in.read(buf);      // Read data into rest of buffer
    buf.flip();        // Flip buffer
    out.write(buf);    // Write header + data to channel
```

This method is often used in conjunction with the `compact` method when transferring data from one place to another.

Returns:

This buffer

rewind

```
public Buffer rewind()
```

Rewinds this buffer. The position is set to zero and the mark is discarded.

Invoke this method before a sequence of channel-write or *get* operations, assuming that the limit has already been set appropriately. For example:

```
    out.write(buf);    // Write remaining data
    buf.rewind();      // Rewind buffer
    buf.get(array);    // Copy data into array
```

Returns:

This buffer

remaining

```
public final int remaining()
```

Returns the number of elements between the current position and the limit.

Returns:

The number of elements remaining in this buffer

hasRemaining

```
public final boolean hasRemaining()
```

Tells whether there are any elements between the current position and the limit.

Returns:

`true` if, and only if, there is at least one element remaining in this buffer

isReadOnly

```
public abstract boolean isReadOnly()
```

Tells whether or not this buffer is read-only.

Returns:

`true` if, and only if, this buffer is read-only

hasArray

```
public abstract boolean hasArray()
```

Tells whether or not this buffer is backed by an accessible array.

If this method returns `true` then the `array` and `arrayOffset` methods may safely be invoked.

Returns:

`true` if, and only if, this buffer is backed by an array and is not read-only

Since:

1.6

array

```
public abstract Object array()
```

Returns the array that backs this buffer *(optional operation)*.

This method is intended to allow array-backed buffers to be passed to native code more efficiently. Concrete subclasses provide more strongly-typed return values for this method.

Modifications to this buffer's content will cause the returned array's content to be modified, and vice versa.

Invoke the `hasArray` method before invoking this method in order to ensure that this buffer has an accessible backing array.

Returns:

The array that backs this buffer

Throws:

`ReadOnlyBufferException` - If this buffer is backed by an array but is read-only

`UnsupportedOperationException` - If this buffer is not backed by an accessible array

Since:

1.6

arrayOffset

```
public abstract int arrayOffset()
```

Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.

If this buffer is backed by an array then buffer position *p* corresponds to array index *p* + `arrayOffset()`.

Invoke the `hasArray` method before invoking this method in order to ensure that this buffer has an accessible backing array.

Returns:

The offset within this buffer's array of the first element of the buffer

Throws:

`ReadOnlyBufferException` - If this buffer is backed by an array but is read-only

`UnsupportedOperationException` - If this buffer is not backed by an accessible array

Since:

1.6

isDirect

```
public abstract boolean isDirect()
```

Tells whether or not this buffer is *direct*.

Returns:

`true` if, and only if, this buffer is direct

Since:

1.6

slice

```
public abstract Buffer slice()
```

Creates a new buffer whose content is a shared subsequence of this buffer's content.

The content of the new buffer will start at this buffer's current position. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's position will be zero, its capacity and its limit will be the number of elements remaining in this buffer, its mark will be undefined. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

The new buffer

Since:

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slice

```
public abstract Buffer slice(int index,
                             int length)
```

Creates a new buffer whose content is a shared subsequence of this buffer's content.

The content of the new buffer will start at position `index` in this buffer, and will contain `length` elements. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's position will be zero, its capacity and its limit will be `length`, its mark will be undefined. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Parameters:

`index` - The position in this buffer at which the content of the new buffer will start; must be non-negative and no larger than `limit()`

`length` - The number of elements the new buffer will contain; must be non-negative and no larger than `limit() - index`

Returns:

The new buffer

Throws:

`IndexOutOfBoundsException` - If `index` is negative or greater than `limit()`, `length` is negative, or `length > limit() - index`

Since:

13

duplicate

```
public abstract Buffer duplicate()
```

Creates a new buffer that shares this buffer's content.

The content of the new buffer will be that of this buffer. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's capacity, limit, position and mark values will be identical to those of this buffer. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

The new buffer

Since:

9

Report a bug or suggest an enhancement
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